

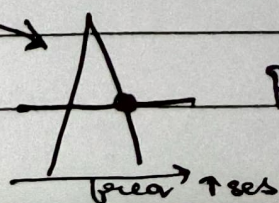
Autonomous Re-enforcement Learning for Resonator Calibration : A Quantum inspired Control system

Qubits are highly volatile and unstable. They are susceptible to noise (especially thermal). So quantum control systems are used to generate microwave pulses to read, & control the state of a qubit as well as input information into the qubit for calculations.

Current methods : $\rightarrow$ Periodic Recalibration

- $\rightarrow$ Manual Calibration based on research
- $\rightarrow$ Brute force : Testing all frequencies (each and every frequency is tested without any history / relation) $\rightarrow$ very sensitive to measurement noise
- $\rightarrow$ PID (proportional Integral Derivative):
 - $\Rightarrow$ A PID controller works as a continuous feedback loop that calculates the "error" b/w desired target & current O/P
 - $\Rightarrow$ Its primary disadvantage is that it struggles with non-linear environment.

lorentzian curve



PID is directionally blind

//_

DQN (Deep Q Network) :

- Provides discrete control while the Lorentzian curve is continuous
- DQN is easier to set up but is slow as it learns from all of memory and causes jitter due to discrete steps

Our Approach :

↗ Smooth tuning & stability

PPO (Proximal Policy Optimization) :

- It is a combination of 2 neural networks that work together
- 1. looks at the current state of circuit (both frequency & amplitude) and decides exactly how much to move the freq
 2. It predicts the rewards for the movement of the first network and maintains the baseline accordingly. It ensures the advantage and that the model is always moving in the right direction towards resonant freq

Proximal → Staying close to previous strategy & prevents the model from making wild jumps

Policy → Policy $\Leftrightarrow$ trained reflexes

It takes 5 I/Ps and O/Ps frequency shift.
It doesn't look up a table but instead feels the ~~cross~~ slope of the curve and reacts accordingly, continuously.

Optimization → Adjusts the network to maximize the reward/gain

5 I/Ps $\Rightarrow$

- $\left. \begin{array}{l} \rightarrow \\ \rightarrow \\ \rightarrow \end{array} \right\} \begin{array}{l} \text{Normalized frequency} \Rightarrow \text{where the agent is currently positioned} \\ \text{Current amplitude (from envelope detector)} \\ \text{Previous amplitude} \end{array}$

$\rightarrow$ Previous action (direction of movement)

$\rightarrow$ Amplitude gradient $\Rightarrow \frac{\Delta \text{amp}}{\Delta \text{freq}}$

what sets our model apart

$\Downarrow$

allows to distinguish b/w amp drop caused by the model / by the env drift

$\left. \begin{array}{l} \text{ppo} \\ \text{advan} \\ \text{-tag} \end{array} \right\}$

$\left. \begin{array}{l} \Delta \text{amp} \\ \Delta \text{freq} \end{array} \right\} \begin{array}{l} \text{enables smooth} \\ \text{tuning \&} \\ \text{stability} \end{array}$

allows to distinguish
b/w amp drop caused by
the model / by the env drift

"fit" the slope of the Lorentzian curve

ensures the value is b/w -1 to 1 to understand the location of model

clip $\left[\frac{\Delta \text{amp}}{\Delta \text{freq} + 1} \times \underbrace{3000}_{\text{scaling factor}}, (-1, 1) \right] \Rightarrow \text{amplitude gradient}$

helps in the differentiation

This one parameter reduced 34% of the mean absolute error / MAE and a near perfect 94.8% of the ^{was} time spent at resonance

avg MAE $\approx 1.5 \text{ KHz}$

lowest MAE $\approx 3670 \text{ Hz}$

500 KHz $\Rightarrow$ resonant frequency

Drift % $\Rightarrow \pm 5\%$

Abs Frequency range $\Rightarrow (475 - 525) \text{ KHz}$

Max Step = 3 KHz (Scaling factor)

depicts both quantum as well as even classical LC System

The reason for using Lorentzian curve is that it is the universal mathematical curve which shows the character of energy absorption and energy dissipation

Ex: black body radiation curve

NOISE REGIMES

➔ To train & test against gradient blindness

Gaussian → random walk
→ lot of drift and can stay at a particular point indefinitely

OU → mean-reverting (oscillates around for itself)
↙ To test and train the model's susceptibility to noise
→ more realistic
→ sudden spikes (due to noise)

Gaussian

Oscstein ^{Uhlenbeck} ~~Uhlenbeck~~

PID

Very Poor → gradient blindness

Moderate → reacts to noise spikes

PQN

Unstable → Not good for non linear systems; lot of jitters

Stable but no precision

PPO

~~X3~~ ➔ Superior →
~~X4~~ ➔ Proper gradient observation and now MAE with 94.8% of the time at resonance

Even better ~~than~~ results with 1300 Hz $\pm$ 95 Hz
MAE & 97% of the time at resonance